	AIR UNIVERSITY
	DEPARTMENT OF COMPUTER ENGINEERING
	EXPERIMENT NO: 2

Lab Tittle: Introduction to Hub, Switch & Router (Switch Administrative Function)

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Objectives: Configure a switch with a name and an IP address. Configure passwords to ensure that access to the CLI is secured. Configure switch port speed and duplex properties for an interface. Save the active configuration. View the switch browser interface

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks:_____ Obtained Mark: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____



DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Introduction to Hub, Switch & Router

Lab Report: 2

Tittle: Introduction to Hub, Switch & Router

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Lab 2

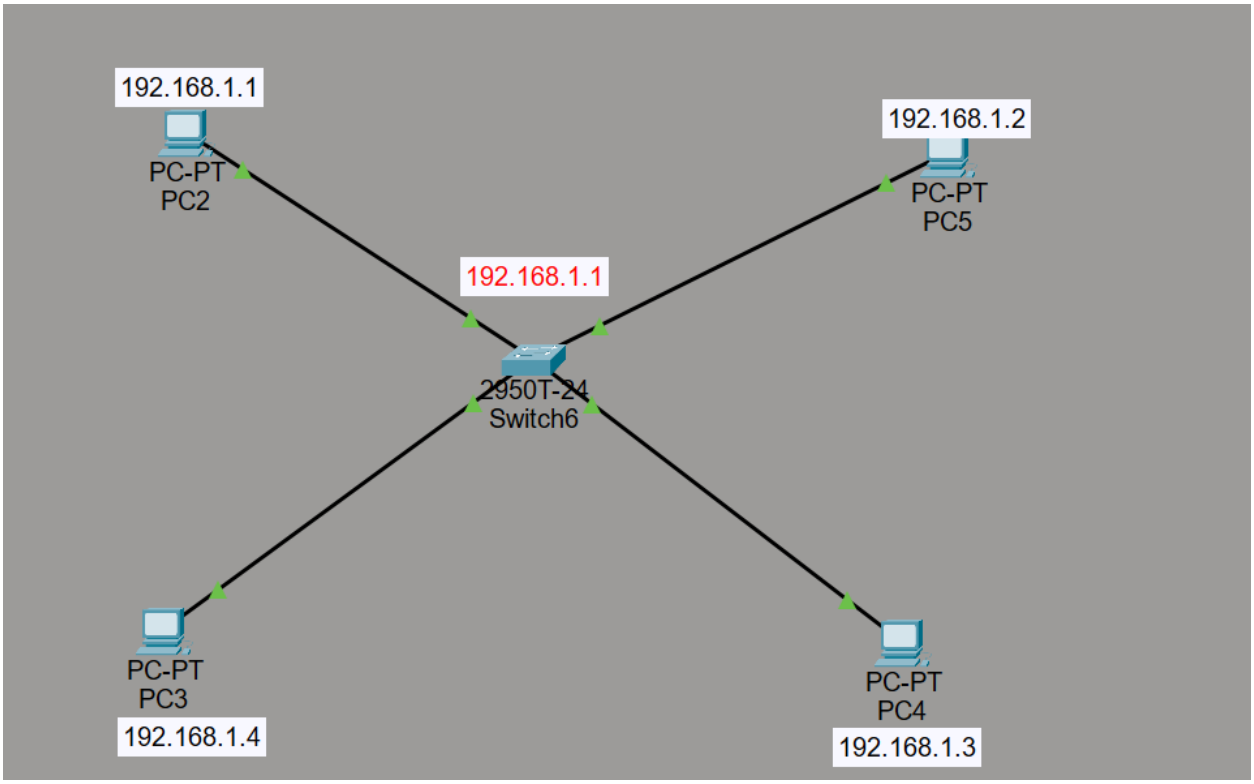
TASK 1:

HUB	SWITCH
1. OSI Layer Operates on the Physical Layer (Layer 1) of the OSI model.	**1. OSI Layer** Operates on the Data Link Layer (Layer 2) of the OSI model.
2. Transmission Type Uses broadcast transmission, sending data packets to all connected ports regardless of destination.	**2. Transmission Type** Supports **unicast, multicast, and broadcast** transmission types, routing traffic only where it needs to go.
3. Port Capacity Typically comes equipped with 4 to 12 ports.	**3. Port Capacity** Offers higher density, typically ranging from **24 to 48 ports**.
4. Collision Domain Contains a single collision domain shared across all connected devices.	**4. Collision Domain** Provides dedicated bandwidth by giving **each port its own separate collision domain**.
5. Transmission Mode Operates strictly in half-duplex mode, meaning devices cannot send and receive data at the same time.	**5. Transmission Mode** Operates in **full-duplex mode**, allowing devices to send and receive data simultaneously.
6. Packet Filtering Does not support packet filtering, as it cannot inspect network traffic or MAC addresses.	**6. Packet Filtering** Provides **packet filtering** by reading and managing MAC addresses to filter and forward data.
7. Intelligence & Device Type Functions as a basic (non-intelligent) **device**, making it cheaper and simpler compared to switches.	**7. Intelligence & Device Type** Functions as an **intelligent device** that directs packets to specific targets, making it more sophisticated and expensive.
8. Speed Capabilities Offers standard speeds starting at **10 Mbps** for original hubs and up to **100 Mbps** for fast hubs.	**8. Speed Capabilities** Delivers high-speed networking, supporting bandwidth speeds from **10**

	Mbps/100 Mbps up to 1 Gbps/10 Gbps and higher.

Layer-2 Device (Data Link Layer)	Layer-3 Device (Network Layer)
• **Primary Function:** Connects devices within the same Local Area Network (LAN) using physical hardware addresses. • **Addressing:** Operates using **MAC (Media Access Control) addresses** to direct traffic. • **Data Unit:** Processes data in the form of **frames**. • **Network Boundary:** Cannot route traffic between different networks or IP subnets; it keeps communication contained inside a single local network. • **Primary Example:** Standard Network Switch.	• **Primary Function:** Connects entirely separate networks together and directs data across different network paths. • **Addressing:** Operates using **IP (Internet Protocol) addresses** to determine routing paths. • **Data Unit:** Processes data in the form of **packets**. • **Network Boundary:** Routers and Layer-3 switches read logical IP destination addresses to forward data across different subnets, LANs, or out to the WAN/Internet. • **Primary Examples:** Router, Layer-3 (Multilayer) Switch.

Lab Task 2



1. **Change name of switch**

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname switch-alpha
switch-alpha(config)#
```

2. **Set enable and secret password**

```
switch-alpha(config)#
switch-alpha(config)#enable password 12345
switch-alpha(config)#enable secret 54321
switch-alpha(config)#
```

3. **Secure VTY lines**

```
switch-alpha(config)#
switch-alpha(config)#line vty 0 15
switch-alpha(config-line)#password 786786
switch-alpha(config-line)#login
switch-alpha(config-line)#exit
switch-alpha(config)#
```

4. **Create VLAN interface & Assign an IP address**

```
switch-alpha(config)#interface vlan 1
switch-alpha(config-if)#ip address 192.168.1.10 255.255.255.0
```

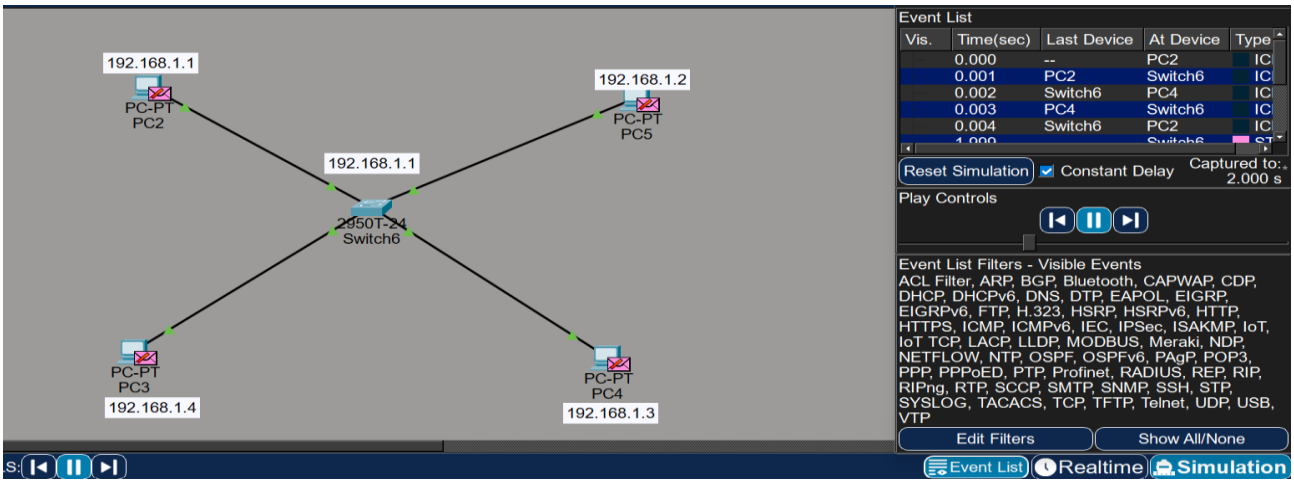
5. **Enable VLAN interface**

```
switch-alpha(config-if)#no shutdown
switch-alpha(config-if)#exit
switch-alpha(config)#exit
switch-alpha#
%SYS-5-CONFIG_I: Configured from console by console
```

6. **Save running configuration as startup configuration**

```
switch-alpha#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
switch-alpha#
```

Ping of pc:



Conclusion:

Hubs are old, simple devices that blindly send data to every connected port and slow down the network. By completing this lab, I successfully connected four computers to a Cisco 2950 switch and secured its command-line interface with passwords. I configured a management VLAN interface with a unique IP address to allow remote access across the network. I also assigned static IP addresses to all connected PCs to put them on the same subnet. Finally, I saved the configuration to the switch's permanent memory so the settings survive a system reboot.

Switches are smart, modern devices that send data directly to the correct destination, making them the standard choice for your lab tasks